

NAME OF THE FACULTY: S. ALIVELU

Lesson Plan: Internet of Things (CM-504) — Academic Year 2025-26

UNIT 1: Introduction of IOT (10 Periods)				
S.No	Date	Name of the topic to be covered	No. Of Periods	completed Date
1	01-12-2025	1.1 INTRODUCTION: Define IOT, Features, Evolution of Connected Devices, Advantages & Disadvantages.	1	
2	03-12-2025	1.1 Components of IoT, Applications of IOT, and various connecting technologies.	1	
3	05-12-2025	1.2 Sensors & Actuators: Need & Features of Sensors, Classification of Sensors; Definition and types of Actuators.	1	
4	08-12-2025	1.3 IoT Architecture: IoT Solutions Architecture, IOT challenges, and IoT Ecosystem.	1	
5	10-12-2025	1.4 Connectivity Technologies: 6LoWPANs - Features, Applications, and Advantages.	1	
6	12-12-2025	1.4 6LoWPANs Technologies: Addressing and different packet formats.	1	
7	15-12-2025	1.4 6LoWPANs protocol stack architecture.	1	
8	17-12-2025	1.4 6LoWPANs Routing protocols (RPL, CORPL, LOADng).	1	
9	19-12-2025	1.4 RFID Technologies: Definition, features, Applications, and Advantages.	1	
10	22-12-2025	1.4 RFID Working principle.	1	
11	24-12-2025	Revision of Unit 1.	1	
(UNIT 1 END)			Total 10 Periods	

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UNIT 2: Data Protocols (14 Periods)				
S.No	Date	Name of the topic to be covered	No. Of Periods	completed Date
1	26-12-2025	2.1 MQTT: Define Message Queue Telemetry Transport (MQTT), list its features.	1	
2	29-12-2025	2.1 Explain MQTT, components, and Methods.	1	
3	30-12-2025	2.1 List and explain Applications of MQTT.	1	
4	02-01-2026	2.1 Define and Explain Secure MQTT (SMQTT).	1	
5	03-01-2026	2.2 CoAP: Define Constrained Application Protocol (CoAP) and its features.	1	
6	05-01-2026	2.2 Explain CoAP and its message types.	1	
7	06-01-2026	2.2 Detailed explanation of CoAP message types.	1	
8	08-01-2026	2.2 Differentiate between COAP and MQTT.	1	
9	19-01-2026	2.3 XMPP: Features of Extensible Messaging and Presence Protocol (XMPP) and its history.	1	
10	20-01-2026	2.3 Explain XMPP and core XMPP Technologies.	1	
11	21-01-2026	2.3 Applications of XMPP.	1	
12	23-01-2026	2.4 AMQP: Features of Advanced Message Queuing Protocol (AMQP).	1	
13	24-01-2026	2.4 Explanation of AMQP in detail.	1	
14	27-01-2026	2.4 Applications of AMQP.	1	
15	28-01-2026	Revision of Unit 2.	1	
		(UNIT 2 END)	Total 14 Periods	

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UNIT 3: Communication Technologies (14 Periods)				
S.No	Date	Name of the topic to be covered	No. Of Periods	completed Date
1	02-02-2026	3.1 IEEE 802.15.4: Features and its preference in IoT communication.	1	
2	03-02-2026	3.1 Explanation of IEEE 802.15.4 protocol.	1	
3	05-02-2026	3.1 IEEE 802.15.4 Variants.	1	
4	06-02-2026	3.1 List and explain IEEE 802.15.4 Types.	1	
5	07-02-2026	3.2 ZIGBEE: Definition, features, and components.	1	
6	09-02-2026	3.2 ZIGBEE different topologies, types, and applications.	1	
7	10-02-2026	3.2 Explanation of different topologies & types of ZIGBEE.	1	
8	11-02-2026	3.3 NFC: Near Field Communication definition, types, and applications.	1	
9	12-02-2026	3.3 Working principle and Modes of operation of NFC.	1	
10	13-02-2026	3.4 Bluetooth: Purpose, features, and functions.	1	
11	14-02-2026	3.4 List applications of Bluetooth.	1	
12	16-02-2026	3.4 Explanation of Bluetooth technology in detail.	1	
13	17-02-2026	3.4 Describing Pico Net.	1	
14	18-02-2026	Revision of Unit 3.	1	
(UNIT 3 END)			Total 14 Periods	

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UNIT 4: Wireless Sensor Networks (14 Periods)				
S.No	Date	Name of the topic to be covered	No. Of Periods	completed Date
1	19-02-2026	4.1 WSN: Importance, Applications, characteristics, and Requirements.	1	
2	20-02-2026	4.1 WSN Network Topologies.	1	
3	21-02-2026	4.2 Sensor Networks: Types - Single Source Single Object & Multiple Object Detection.	1	
4	23-02-2026	4.2 Sensor Networks: Types - Multiple Source Single Object & Multiple Object Detection.	1	
5	24-02-2026	4.3 Challenges in WSN: Node definition, and Node Behavior in WSNs.	1	
6	25-02-2026	4.3 Security Services in WSN.	1	
7	26-02-2026	4.3 Information theoretic self-management in WSN.	1	
8	27-02-2026	4.4 Applications of WSN: Wireless Multimedia Sensor Networks (WMSN).	1	
9	28-02-2026	4.4 Stationary Wireless Sensor Networks.	1	
10	02-03-2026	4.4 Mobile Wireless Sensor Networks.	1	
11	03-03-2026	4.4 M2M: Machine to Machine Communications applications and features.	1	
12	04-03-2026	4.4 M2M sensor nodes.	1	
13	05-03-2026	Revision of WSN concepts.	1	
14	06-03-2026	Revision of WSN applications.	1	
(UNIT 4 END)			Total 14 Periods	

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UNIT 5: Role Of IOT (8 Periods)				
S.No	Date	Name of the topic to be covered	No. Of Periods	completed Date
1	07-03-2026	5.1 Role of IOT in automation: importance, applications, advantages & disadvantages.	1	
2	09-03-2026	5.1 Impact of IOT on Industrial Automation & Types of Industrial Automation.	1	
3	10-03-2026	5.1 Role of IOT in automation application: Health care.	1	
4	11-03-2026	5.1 Role of IOT in automation application: Smart Home.	1	
5	12-03-2026	5.1 Role of IOT in automation applications: Smart Cities & Smart classrooms.	1	
6	13-03-2026	5.1 Role of IOT in automation application: Smart Energy.	1	
7	14-03-2026	5.1 Role of IOT in automation application: Smart Transportation and Mobility.	1	
8	16-03-2026	5.1 Role of IOT in automation application: Smart Factory & Syllabus Completion.	1	
9	17-03-2026	Revision of Unit 5.	1	
(UNIT 5 END)			Total 8 Periods	

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